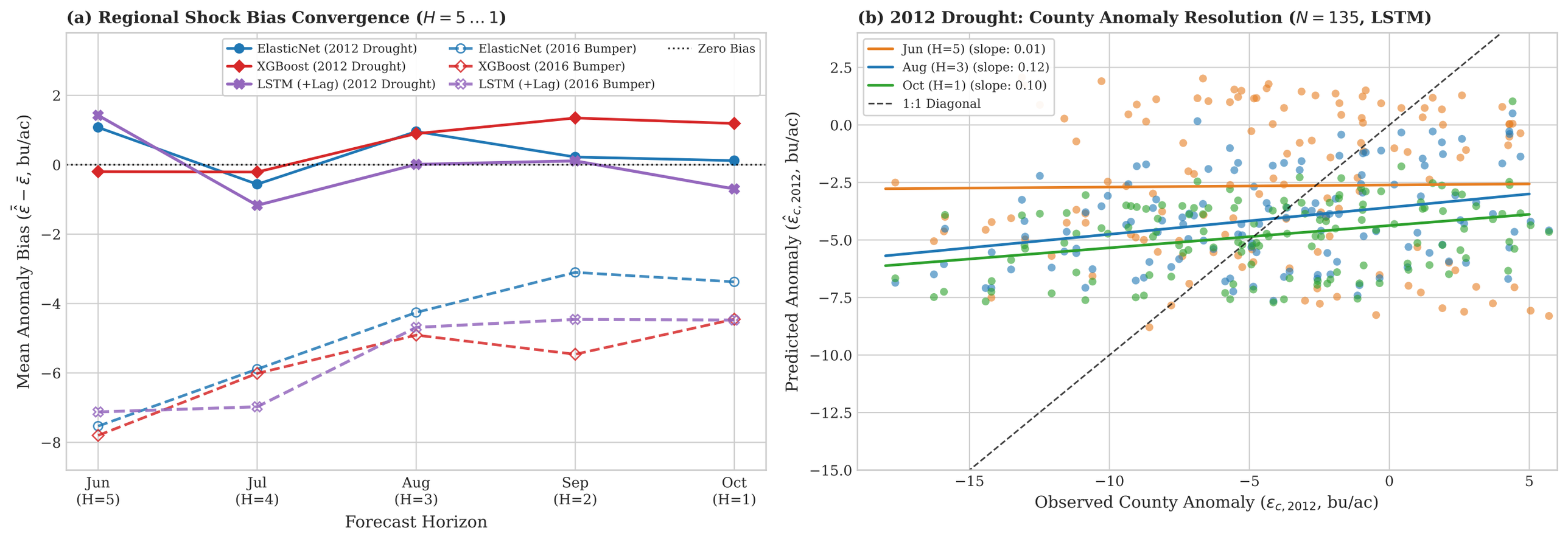
(a) Regional Shock Bias Convergence ($H = 5 \ldots 1$)
ElasticNet (2012 Drought)
XGBoost (2012 Drought)
LSTM (+Lag) (2012 Drought)
ElasticNet (2016 Bumper)
XGBoost (2016 Bumper)
LSTM (+Lag) (2016 Bumper)
Zero Bias
Mean Anomaly Bias ($\bar{\hat{\varepsilon}} - \bar{\varepsilon}$, bu/ac)
2
0
−2
−4
−6
−8
Jun (H=5)
Jul (H=4)
Aug (H=3)
Sep (H=2)
Oct (H=1)
Forecast Horizon
(b) 2012 Drought: County Anomaly Resolution ($N = 135$, LSTM)
Jun (H=5) (slope: 0.01)
Aug (H=3) (slope: 0.12)
Oct (H=1) (slope: 0.10)
1:1 Diagonal
Predicted Anomaly ($\hat{\varepsilon}_{c,2012}$, bu/ac)
2.5
0.0
−2.5
−5.0
−7.5
−10.0
−12.5
−15.0
−15
−10
−5
0
5
Observed County Anomaly ($\varepsilon_{c,2012}$, bu/ac)